Year 10 Science
Physical Sciences
Worksheet - Speed and Velocity

SPEED

The speed of a body is the rate of change of distance by the body.

The average speed of a body over a distance travelled in a time interval can be determined by:

$$\text{Speed (av)} = \frac{d}{t}$$

where d = distance (measured from zero)
t = time interval (measured from zero)

Speed is a scalar quantity as it involves no specific direction. Distance is the total length of the path covered by the body.

If to accurately describe motion direction is required, then *VELOCITY* is used in place of speed.

VELOCITY

The velocity of a body is the rate of change of displacement of the body.

The average velocity of a body over a displacement in a measured time interval can be determined by:

$$v_{av} = \frac{s}{t}$$

v_{av} = average velocity
s = displacement (measured from zero)
t = time interval (measured from zero)

Velocity is a vector quantity. *The direction of the velocity is that of the displacement (unless otherwise stated).*

UNITS OF VELOCITY AND SPEED

Velocity and speed are both determined by dividing units of length (m) by units of time (s).

Units: $\frac{\text{metres}}{\text{seconds}} = \frac{m}{s} = m\ s^{-1}$ (or m/s)

DETERMINATION OF DISPLACEMENT

The equation for average velocity can be rearranged to determine the displacement.

$$v_{av} = \frac{s}{t}$$

rearranging gives: $s = v_{av} \times t$

similarly $d = \text{speed (av)} \times t$

TYPE EXAMPLE 7: Calculate the average velocity of a vehicle which travels 3 km east in 5 minutes.

$v_{av} = ?$
$s = 3\ km = (3 \times 1\,000)\ m$
$t = 5\ minutes = (5 \times 60)\ s$

$$v_{av} = \frac{s}{t} = \frac{3 \times 1\,000}{5 \times 60} = 10\ m\ s^{-1}$$

i.e. the average velocity is 10 m s^{-1} east.

TYPE EXAMPLE 8: Determine the displacement of a body which moves with a uniform velocity of 2.4 m s^{-1} for 50 s.

$v_{av} = 2.4 \text{ m s}^{-1}$
$s = ?$
$t = 50 \text{ s}$

$$v_{av} = \frac{s}{t}$$
$$\therefore \quad s = v_{av} \times t$$
$$= 2.4 \times 50$$
$$= 120 \text{ m}$$

i.e. the displacement is 120 m in the direction of the velocity.

SET 6: VELOCITY I

1. What is the average speed of a sprinter who runs 100 m in 12.5 s?
2. A car takes 80 s to complete one lap of a 2 km circuit. What is its average speed (in m s^{-1})?
3. A body travels 36 m in 4 s. What is its average speed?
4. How far will a bird fly in 25 s travelling with an average speed of 6 m s^{-1}?
5. A person swims from one end of a pool to the other at an average velocity of 1.5 m s^{-1} in 20 s. What is the length of the pool?
6. Determine the distance travelled by a plane in 1 hour if it flies due south with a uniform velocity of 90 m s^{-1}
7. How long will it take for a body moving with a uniform velocity of 7 m s^{-1} to cover 98 m?
8. A cricket ball is thrown a distance of 90 m from the boundary to the wicket at an average horizontal speed of 30 m s^{-1}. How long will it take for the ball to reach the wicket?
9. Determine the time taken for a horse running with an average speed of 14 m s^{-1} to complete one circuit of a 840 m track.
10. How long will it take to complete a trip from Perth to Kalgoorlie, a distance of 595 km, travelling at an average speed of 70 km h^{-1}?
11. A train moving with an average speed of 80 km h^{-1} takes 0.25 hours to travel from one railway siding to the next. What is the distance between the two sidings?
12. Determine the average speed (km h^{-1}) of a plane which flies a distance of 720 km in 1.5 hours?

SET 6

1. 8 m s^{-1} **2.** 25 m s^{-1} **3.** 9 m s^{-1} **4.** 150 m **5.** 30 m
6. 324 000 m **7.** 14 s **8.** 3 s **9.** 60 s **10.** 8.5 h
11. 20 km **12.** 480 km h^{-1}